

**STATE GDP FORECASTING WITH MACHINE LEARNING:
A CASE STUDY OF RIO GRANDE DO SUL***
**PREVISÃO DO PIB ESTADUAL COM APRENDIZADO DE MÁQUINA:
UM ESTUDO DE CASO DO RIO GRANDE DO SUL**

Eduarda Teixeira Sayago**
Marcelo Savino Portugal***

ABSTRACT

The forecasting of GDP is essential for policy-makers and businesses to guide decisions, including at the state level. This study evaluates the efficacy of the machine learning methods LASSO, Elastic Net and Random Forest in improving the forecasting of Rio Grande do Sul's GDP, relative to a SARIMA model. Using a rolling window approach, 1-step-ahead (short) and 1-year-ahead (long) horizons are assessed at monthly and quarterly frequencies. Results show that LASSO obtained the lowest short-term errors, while Random Forest is most accurate long-term. The DM test indicates machine learning significantly improve forecasts against the benchmark in most horizons, endorsing ML as a valuable tool for regional GDP prediction.

Keywords: GDP Forecasting, Machine Learning, LASSO, Elastic Net, Random Forest, SARIMA.

RESUMO

A previsão do PIB é essencial para que formuladores de políticas e empresas orientem suas decisões, inclusive em nível estadual. Este estudo avalia a eficácia dos métodos de aprendizado de máquina LASSO, Elastic Net e Random Forest em aprimorar a projeção do PIB do Rio Grande do Sul, em comparação a um modelo SARIMA. Utilizando janelas móveis, são analisados horizontes de 1 passo à frente (curto prazo) e de 1 ano à frente (longo prazo), nas frequências mensal e trimestral. Os resultados mostram que o LASSO apresentou os menores erros de curto prazo, enquanto o Random Forest foi o mais preciso no longo prazo. O teste DM indica que o aprendizado de máquina melhora significativamente as previsões contra o benchmark na maioria dos horizontes, endossando-o como uma ferramenta valiosa para a projeção do PIB regional.

Palavras-chave: Previsão do PIB, Aprendizado de Máquina, LASSO, Elastic Net, Random Forest, SARIMA.

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**Undergraduate student in Economics at UFRGS. (eduarda.sayago@gmail.com).

***Advisor. PhD in Economics from the University of Warwick (1992). Professor in the Department of Economics and International Relations at UFRGS. (msp@ufrgs.br).

1 INTRODUCTION

Gross domestic product (GDP) forecasting remains highly relevant in macroeconomic analysis, also at the regional level. For fiscal governance, tax authorities and policymakers rely on these projections to estimate future revenue streams, enabling governors to organize state budgets with precision and determine appropriate public investments. In the private sector, these forecasts are equally critical, allowing firms to calibrate current production levels against anticipated demand, minimizing the risks associated with inventory surpluses or supply shortages. They are also used in financial markets, where expectations about GDP growth can influence stock valuations and portfolio strategies.

Therefore, improving the accuracy of the state’s GDP forecasts enables policymakers, firms and investors to better anticipate economic cycles, allocate resources more efficiently, and design targeted interventions in response to shocks. The task becomes even more crucial in periods of elevated uncertainty, such as the post-pandemic era following Covid-19 (Sabri *et al.*, 2023) or commodity and climate shocks, such as droughts (Tonetto; Pique; Fochezatto, *et al.*, 2024). This study focuses on whether machine learning models can improve forecasts of Rio Grande do Sul’s GDP, a Brazilian state, relative to a standard econometric benchmark, at two frequencies (monthly and quarterly).

Traditional econometric models, such as the Seasonal Autoregressive Integrated Moving Average (SARIMA), have been widely used for forecasting of GDP and other economic variables, given their ability to model time series data with seasonality (Gu *et al.*, 2022). However, they often rest on a set of strong assumptions, such as linearity and stationarity, that can make them rigid and unable to capture the complexity of real-world data (Oancea; Simionescu, 2024; Pinto, 2011). These limitations became starker with the advent of big data, requiring more flexible models in regard to data structure.

Consequently, machine learning techniques emerged as a powerful tool for GDP prediction in settings with complex and multi-source datasets (Masini; Medeiros; Mendes, 2023). Since ML methods rely on data-driven methods for variable selection, they are less affected by high dimensionality (Athey; Imbens, 2017), and show potential for their ability to capture nonlinear relationships between variables (Izenman, 2008). The availability of diverse data sources can strengthen the prediction accuracy of economic variables, as demonstrated by Lindenmeyer, Skorin, and Torrent (2021), whose boosting ML method used more than eight hundred predictors to forecast energy demand and resulted in a better fit to the target variable than the SARIMA benchmark.

Several studies endorse the potential of ML on improving forecasting accuracy. For instance, K. Safi (2016) found that an artificial neural network (ANN) reduced RMSE relative to ARIMA for the GDP of Palestine, and Flannery (2020) show Random Forest produced strong quarterly GDP forecasts for Ireland. Large comparative exercises reinforce these findings: Medeiros, Vasconcelos, *et al.* (2021) compared 26 methods (including LASSO, Elastic Net, factor and ensemble approaches) and documented accuracy gains of up to 30% for some ML methods versus standard benchmarks. Studies focused on national and subnational variables likewise report the robustness of ensemble approaches (Araujo; Gaglianone, 2023; Yang *et al.*, 2024).

Meanwhile, ML methods like the least absolute shrinkage and selection operator (LASSO) and the Elastic Net remain valuable in many forecasting contexts, producing interpretable and parsimonious models even in high dimensionality settings (Athey; Imbens, 2017). Empirical work shows these methods frequently beat simple autoregressive benchmarks, for example, Cepni, Güney, and Swanson (2019) and Medeiros and Vasconcelos

(2016) reported gains in the prediction of GDP and other macroeconomic variables for the LASSO when mixing monthly and quarterly data or using a rolling window evaluation.

Despite the broad literature on GDP forecasting globally and the existing studies focused on the Brazilian economy, research that applies machine learning models for subnational GDP forecast remains scarce. Similar papers using ML at the regional level include Portugal and Fernandes (1996) for predicting Rio Grande do Sul’s monthly industrial production and Lindenmeyer, Skorin, and Torrent (2021), for energy demand. Baggio (2017) estimates and nowcasts RS’s GDP with a State-Space model, but does not explore ML approaches, and Carneiro Júnior and Souza (2024) does so for predicting economic activity of Mato Grosso do Sul, another Brazilian state.

Rio Grande do Sul is currently the 5th biggest state economy of Brazil, representing 6.02% of the national GDP. Agribusiness and its associated industries represent a substantial share of the state’s economic activity, with its temperate climate and extensive plains favoring grain cultivation, viticulture, and cattle raising. Because of these characteristics, predicting Rio Grande do Sul’s GDP requires considering certain caveats. The economy has strong cyclical factors related to agribusiness, being not only sensitive to commodity price fluctuations in the international market, but also strongly influenced by climatic conditions, as floods, droughts, and occasionally frosts are especially deleterious to certain crops and to pasture quality for livestock, directly affecting output growth.

In addition, the state’s GDP time series shows a structural break in the second half of the sample, with growth occurring at a significantly different pace. These factors increase the volatility of Rio Grande do Sul’s economic performance relative to the national series, making it more difficult for forecasting models to adapt to, but also more interesting from a theoretical standpoint. This setting is particularly suitable for testing machine learning models, given their handling of nonlinearity.

In this light, the present study investigates whether machine learning methods deliver more accurate out-of-sample forecasts for the GDP of Rio Grande do Sul compared to traditional econometric models. The models LASSO, Elastic Net and Random Forest are evaluated against a SARIMA benchmark, with forecasts in the short-term (one-step-ahead) and long-term (one-year-ahead) horizons, for two time frequencies: monthly and quarterly. Out-of-sample forecasts are generated using a rolling window approach, which is suitable for dealing with the structural changes in the series. Models are evaluated through RMSE and MAPE errors, as well as a Diebold-Mariano test.

The empirical contribution of this exercise comes in two ways. First, it develops predictive models for the forecasting of GDP, incorporating macroeconomic and weather data. Second, it fills an important gap by generating insights on the use of machine learning to predict the regional economy of Rio Grande do Sul. By doing so at two different frequencies, complementary environments are explored: monthly data contain higher frequency indicators that provide more information to the models, while quarterly data provide a scarcer environment and align with official GDP releases.

This study is organized into five main sections. Section 2 addresses forecasting strategies, models employed and evaluation metrics used to assess predictive accuracy. Section 3 focuses on dataset details and pre-processing techniques, while Section 4 presents and discusses the results obtained in the monthly and quarterly exercises. Finally, Section 5 concludes the work by summarizing the most important details and findings.

2 METHODOLOGY

This section presents the methodological techniques employed in the forecasting exercise¹, and explains in more detail the machine learning methods that were applied in this study, followed by the benchmark model, SARIMA. Finally, the evaluation metrics are introduced, as well as the significance test of Diebold-Mariano.

2.1 FORECASTING STRATEGIES

Following Medeiros, Vasconcelos, *et al.* (2021) and Boesch and Ziegelmann (2025), in this work it was employed a direct forecast approach, where the target variable h periods ahead of t , y_{t+h} , is modeled with the set of predictors up to time t , with distinct datasets used for each forecast horizon.

The chosen horizons for the monthly series were 1 and 12 periods ahead, while the quarterly series horizons were 1 and 4, with the purpose of capturing predictions one-step-ahead and one-year-ahead, the most important periods for decision-makers. To account for the temporal structure of the data, a rolling window scheme was used, where the models are refitted in windows of fixed length, and generate the out-of-sample observations. The window size is dependent on the forecasting horizon h and the number of lagged variables used in the model. All models are evaluated based on a fixed number of point forecasts n , that corresponds to 30% of the dataset².

2.2 METHODOLOGY

The shrinkage methods LASSO and Elastic Net were chosen for their ability to reduce the dimensionality of the covariate space, whereas the ensemble method Random Forest is specially useful for capturing nonlinear relationships even in the high dimensionality framework. The traditional SARIMA model serves as a benchmark, due to its extensive application in previous time series forecasting research.

The objective is to produce an h -step ahead forecast of the target variable y_{t+h} from the set of predictors $\mathbf{x}_t = (x_{t,1}, \dots, x_{t,p})$, which is based on information up to time t . A general model can describe their relationship as

$$y_{t+h} = f_h(\mathbf{x}_t) + \epsilon_{t+h}, \quad (1)$$

where ϵ_{t+h} is the prediction error and $f_h(\mathbf{x}_t)$ represents the function that connects the predictors in the set $\mathbf{x}_t$ to the future response y_{t+h} .

In this work two levels of flexibility are assumed for $f_h(\cdot)$:

1. very flexible, without any shape constraints, leading to a nonparametric model;
2. linear on the predictors, leading to a semiparametric model, since it is not assumed a specific distribution for ϵ_{t+h} .

For methods such as the Random Forest, the function $f_h(\cdot)$ is not defined a priori, which means that its overall shape, apart from being constant by steps, will be completely chosen by the interaction between the estimation method and the data. On the other hand,

¹In this work, AI-based chat bot tools were used to assist with grammar checking and debugging code in the empirical analysis.

²The exercise was executed through the R software (version 4.2.2). The code, packages and data used in this article are available at github.com/eduarda-sayago/gdp-forecasting-with-ML.

methods such as LASSO and Elastic Net are usually applied upon a linear model, that is, $f_h(\cdot)$ is a linear combination of the predictors, as in

$$y_{t+h} = \beta_0 + \beta_1 x_{t,1} + \dots + \beta_p x_{t,p} + \epsilon_{t+h}. \quad (2)$$

2.2.1 Least Absolute Shrinkage and Selection Operator

The least absolute shrinkage and selection operator, or LASSO, originally presented by Tibshirani (1996), was developed to address some limitations of Ordinary Least Squares (OLS) regression, particularly its high variance and lack of interpretability in high-dimensional settings. The LASSO is one of the most influential methods in the family of penalized regression methods (Athey; Imbens, 2017), and, like the OLS regression, minimizes the residual sum of squares (RSS). In a setting where the dataset contains n observations of p predictors, with each observation n in the i -th row of the j -th column, the RSS is defined as

$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2. \quad (3)$$

However, LASSO introduces an additional constraint to improve model performance, minimizing not only the RSS, but also the l_1 norm of the coefficient vector, defined as the sum of the absolute values of the regression coefficients, and regulated by a constant λ . The objective function for LASSO can be written as

$$\min_{\beta_1, \dots, \beta_p} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|, \quad (4)$$

where β_j represents the coefficient for the j -th predictor variable. The penalty causes the coefficient estimates to shrink toward zero. When $\lambda = 0$, the penalty term disappears, and LASSO becomes equivalent to OLS. As λ increases, the penalty forces more coefficients to shrink, and potentially become exactly zero. Such shrinkage helps reduce the variance of the method, improving its predictive performance. The potential to reduce coefficient estimates to null is what allows LASSO to perform variable selection, a key advantage over similar penalized regression models, which makes the resulting model more interpretable, as it retains only the most relevant predictors and discards those that do not contribute significantly to the prediction task.

2.2.2 Elastic Net

The Elastic Net (ENet) was first proposed by Zou and Hastie (2005) and is a blend of LASSO and Ridge Regression techniques, usually used in linear models. Whereas the LASSO performs variable selection by shrinking coefficients towards zero, Ridge regression encourages more evenly distributed coefficients among correlated variables, averaging their effects rather than eliminating them. The Elastic Net addresses the limitations of using LASSO or Ridge alone by combining their strengths, allowing for both variable selection and stable handling of correlated predictors (Hastie; Tibshirani; Friedman, 2009).

The Elastic Net penalty combines the LASSO and Ridge penalties, as shown in Table 1. The ridge component (second term) promotes the averaging of correlated features, while the LASSO component (first term) promotes sparsity by selecting a subset

Table 1 – Penalty functions

Estimator	Penalty: $p(\beta_1, \dots, \beta_p; \lambda)$
Ridge	$\lambda \sum_{j=1}^p \beta_j^2$
LASSO	$\lambda \sum_{j=1}^p \beta_j $
ENet	$\lambda \left[\alpha \sum_{j=1}^p \beta_j + (1 - \alpha) \sum_{j=1}^p \beta_j^2 \right]$

Source: Boesch and Ziegelmann (2025).

of those averaged coefficients. The balance between these two behaviors is regulated by the parameter α .

2.2.3 Random Forest

Random Forest is an ensemble machine learning method that builds upon the strengths of decision trees while reducing their weaknesses. Decision trees partition the feature space into regions (or “leaves”) containing similar observations and make predictions based on the average target value within each region. While a single decision tree usually suffers from high variance and limited out-of-sample accuracy, a combination of many trees can become a strong predictor.

The Random Forest, introduced by Breiman (2001), is a variation of the Bagging³, where it uses many trees and averages their predictions similarly, but differs by implementing more variability between trees, performing each tree split with a random subset of features. Each tree (b) in the forest defines an estimator $\hat{f}_b(x_t)$ of the unknown predictive function $f_h(x_t)$ using a regression tree fitted on a bootstrap sample. The Random Forest estimator of f_h is the average function

$$\hat{y}_{t+h} = \frac{1}{B} \sum_{b=1}^B \hat{f}_{b,h}(\mathbf{x}_t), \quad (5)$$

where the forecasted value $\hat{y}_{t+h}$ is obtained as the average of the B base learners’ predictions. The lower correlation between the trees, acquired by the random subset of variables at each split (*mtry*), leads to more stable predictions, reducing the variance of individual trees and improving results.

2.2.4 Seasonal Autoregressive Integrated Moving Average

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is an extension of the ARIMA model, designed to account for the presence of seasonality. The characteristics of the ARIMA in detecting non-seasonal trends and autocorrelation structures are kept, while incorporating the capture of recurring patterns in the time series data⁴ (Hyndman; Athanasopoulos, 2018). The structure of a SARIMA model can

³Bagging (Bootstrap Aggregating) is an ensemble technique that builds multiple models on bootstrap resamples of the data and averages their predictions to reduce variance (Breiman, 1996).

⁴Due to the scale of the exercise, the *auto.arima()* function from the *forecast* R package was used to automatically fit the best model to each window. The parameters were adjusted to allow the algorithm to also consider non-seasonal models, with model selection based on the Bayesian Information Criterion (BIC).

be defined as

$$\text{ARIMA} \quad \underbrace{(p, d, q)}_{\text{Non-seasonal component}} \quad \times \quad \underbrace{(P, D, Q)_m}_{\text{Seasonal component}}, \quad (6)$$

where m is the seasonal period, p is the autoregressive order, q is the moving average order and d is the degree of first differencing. The terms in the seasonal part of the model are similar, but involve backshifts of the seasonal period. An $\text{ARIMA}(1,1,1)(1,1,1)_m$ can be represented in backshift notation as

$$(1 - \phi_1 B)(1 - \Phi_1 B^m)(1 - B)(1 - B^m)y_t = (1 + \theta_1 B)(1 + \Theta_1 B^m)\epsilon_t. \quad (7)$$

2.3 PERFORMANCE METRICS

The evaluation and comparison of forecast accuracy for each horizon h were carried out using the root mean squared error (RMSE) and the mean absolute percentage error (MAPE), similarly to Yoon (2021) and Medeiros, Vasconcelos, *et al.* (2021). They are defined as follows:

$$\text{RMSE}(y, \hat{y}) = \sqrt{\sum_{t=1}^n \frac{(y_{t+h} - \hat{y}_{t+h})^2}{n}} \quad (8)$$

$$\text{MAPE}(y, \hat{y}) = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{y_{t+h} - \hat{y}_{t+h}}{y_{t+h}} \right|. \quad (9)$$

All metrics are calculated over the difference between the observed values of the series, y_t , and the predicted values $\hat{y}_t$ from each model, where $t = 1, \dots, n$, h is the horizon and n is the last 30% of the total number of observations T in the dataset.

For easier visualization and inspired by the Ratio RMSE used in Medeiros, Vasconcelos, *et al.* (2021), the tables are exposed with the relative measure of RMSE, against the error metrics from the benchmark, SARIMA, as it shows:

$$r\text{RMSE}_h^m = \frac{\text{RMSE}_h^{\text{model}}}{\text{RMSE}_h^{\text{bench}}}. \quad (10)$$

With these measures, if $r\text{RMSE} < 1$, the model m had a better performance in that metric, for horizon h , against the SARIMA (*bench*). Following Huang and Zhou (2017), Caldeira, Moura, and Santos (2016), and Lv and Wu (2022), the analysis was complemented with the cumulative squared forecast error (CSFE) metric, originally presented by Welch and Goyal (2008).

The CSFE accumulates, over time, the differences between the squared errors of the benchmark and those of the model. Caldeira, Moura, and Santos (2016) highlights the CSFE as a useful tool for comparing performance across the sample, as it allows for a clearer identification of where the model makes its largest errors. Using the same notation as in previous studies, the CSFE can be formally expressed as

$$\text{CSFE}_{h(m)} = \sum_{t=1}^n \left[\left(y_{t+h} - \hat{y}_{t+h|\text{benchmark}} \right)^2 - \left(y_{t+h} - \hat{y}_{t+h|m} \right)^2 \right], \quad (11)$$

where $\hat{y}_{t+h}$ denotes the realized value of the variable of interest at horizon h , $\hat{y}_{t+h|\text{benchmark}}$ is the h -step-ahead forecast generated by the benchmark model, and $\hat{y}_{t+h|m}$ is the

corresponding forecast from model m . A positive value of $\text{CSFE}_{h(m)}$ indicates that model m outperforms the benchmark, or, it produces smaller cumulative squared forecast errors, while a negative value suggests inferior predictive performance.

2.4 SIGNIFICANCE TEST

For evaluating whether the benchmark and the competing forecasting model show a significant difference in predictive accuracy, the Diebold-Mariano (DM) test was applied (Diebold; Mariano, 2002). It is a statistical test that compares the forecast errors from the two models over a common evaluation sample to determine if one model consistently outperforms the other. To formalize this comparison, the test computes a loss differential between the two models, defined as

$$d_t = g(e_{1,t}) - g(e_{2,t}), \quad (12)$$

where $e_{1,t}$ and $e_{2,t}$ denote the forecast errors from the models 1 and 2 for period t , respectively, and $g(\cdot)$ represents a loss function, such as $g(e_t) = e_t^2$. The DM evaluates the null hypothesis of equality between the terms, or $H_0 : E(d_t) = 0$. The test statistic is computed as follows:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}, \quad (13)$$

where $\bar{d}$ is the sample mean of d_t and $\widehat{\text{Var}}(\bar{d})$ is a heteroskedasticity- and autocorrelation-consistent estimate of its variance.⁵

⁵For the Diebold–Mariano (DM) test, the function *dm.test()* from the R package *forecast* was used. The loss function was the default squared forecast error, the null hypothesis was tested against a two-sided alternative, and the long-run variance was estimated using a Bartlett kernel to account for possible autocorrelation in the forecast errors.

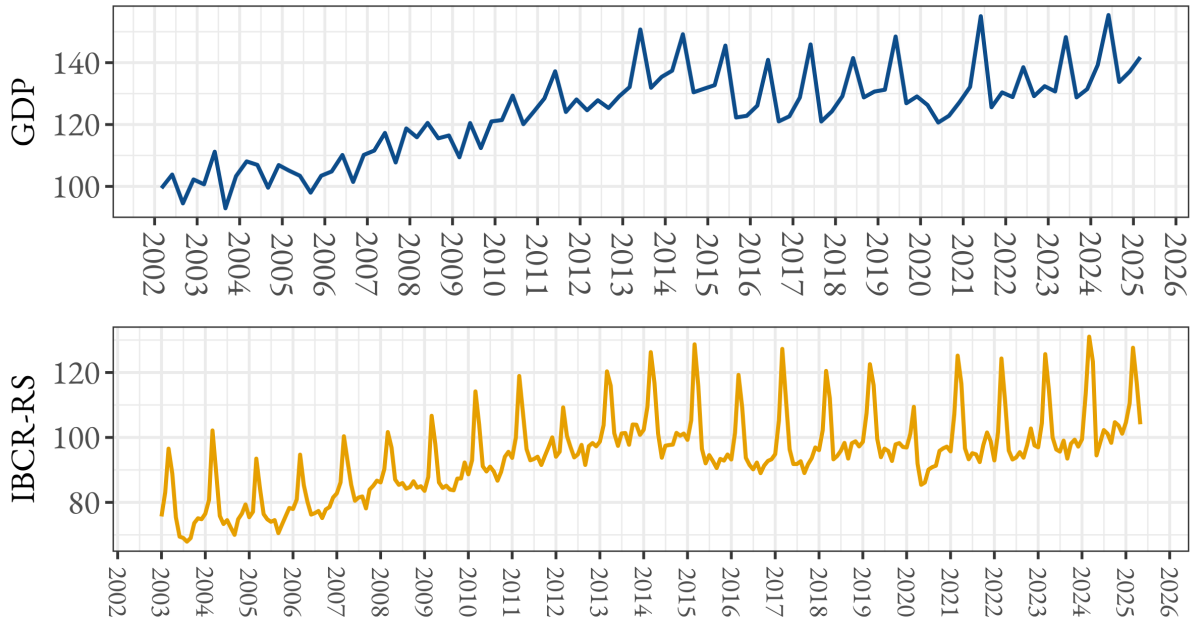
3 DATA TREATMENT

This section presents details on the dataset utilized in the exercise and the data transformations applied in the monthly and quarterly settings.

3.1 DATASET

For the monthly analysis, economic activity in Rio Grande do Sul is represented by the Regional Economic Activity Index (IBCR-Rs) from the Brazilian Central Bank's Time Series Management System. For the quarterly analysis, the dependent variable is the GDP of Rio Grande do Sul, obtained from Departamento de Economia e Estatística (DEE). Both are expressed in index form.

Figure 1 – GDP series of Rio Grande do Sul, in quarterly and monthly frequency. Series expressed in index form (Q1 2002 = 100 for GDP, 01/2022 = 100 for IBCR-Rs).



Source: Quarterly GDP data from Department of Economics and Statistics (DEE); monthly proxy (IBCR-RS) from the Central Bank of Brazil.

The monthly dataset covers January 2003 to June 2025, encompassing 257 observations, while the quarterly dataset goes from Q1 2002 to Q1 2025, with 93 observations. The set of predictors includes 231 features at the subnational (166, including weather), national (58), and international (7) levels. For the monthly exercise, the same variables were used, except for purely quarterly series (e.g., Brazilian GDP), resulting in 217 features. The set of predictor variables includes a range of economic and climate time series, indicated in Table 2:

Table 2 – Summary of dataset and variables (Rio Grande do Sul)

Category	Quantity
Real Economy	8
Price Indexes and Components	13
Industry	15
Agribusiness	10
Services and Trade	4
External Sector	5
Money Supply	6
Interest Rates	6
Commodities	7
Financial Market	3
Public Sector	6
Uncertainty Indicators	3
Climate*	143

Source: author’s own elaboration.

Notes: *11 variables chosen for the 13 most economically significant municipalities of the state⁶.

Four lags of each predictor were incorporated to capture dynamic effects, and seasonal dummies were added for both frequencies. Outlier dummies were included for events such as the Covid-19 pandemic and the 2013 structural trend shift suggested by Figure 1. Meanwhile, as no apparent change can be seen in Q2 2024, when extreme floods impacted the state, a dummy for the event was discarded. It is possible that the floods impacted certain municipalities individually, but had little effect on the state’s economy as a whole (Tonetto; Pique; Rapetti, *et al.*, 2025).

3.2 DATA PROCESSING

To prepare the data for the subsequent analytical procedures, a series of processing steps was implemented to address missing values and resample variables for the appropriate time frequency. Missing values in the meteorological data provided by the Instituto Nacional de Meteorologia (INMET) were handled through time series interpolation, using the function *na.interp* from the *forecast* R package, which in case of a seasonal series, computes a STL decomposition, performs linear interpolation and adds back the seasonal component. For resampling, daily price series were shifted to monthly by taking the last value of each period. Finally, for the quarterly analysis, monthly series were aggregated to quarterly values by calculating the mean of each three-month period.

Before fitting the model in the monthly analysis, all series were inspected and, when appropriate, transformed using logarithms to reduce skewness and stabilize variance. Following the process used in Lindenmeyer, Skorin, and Torrent (2021), for series that were strictly positive, a standard natural logarithm was taken. When the series contained zeroes but no negative values, a small constant was added before taking logs, so that there would be no undefined values at zero. Variables that included only negative values were left untransformed, and if it contained both negative and zero values, the series was shifted

⁶The municipalities were chosen for their economic importance and data availability (INMET). The list comprises Bagé (RS), Bom Jesus (RS), Caxias do Sul (RS), Cruz Alta (RS), Encruzilhada do Sul (RS), Passo Fundo (RS), Pelotas (RS), Porto Alegre (RS), Santa Maria (RS), Santa Vitória do Palmar (RS), São Luiz Gonzaga (RS), Torres (RS) and Uruguaiana (RS).

upward by one unit⁷. In the quarterly setting, the logarithm transformations strongly increased MAPE error.

For each model, the tuning parameters were selected through 5-fold-cross-validation (CV) and chosen according to the smallest RMSE. In LASSO, the regularization parameter λ was chosen via the R package *glmnet*, which performs CV with the function *cv.glmnet*. For the Elastic Net, both λ and the mixing parameter α (that determines the balance between LASSO and Ridge penalization) were tuned simultaneously using a grid search over a broad set of candidate values via the R package *caret*. As for Random Forest, the parameter *mtry*, which defines the number of variables randomly selected at each split, was tested over a small set of possible values, commonly recommended in the literature⁸. This process ensured that each model used the tuning configuration that performed best on unseen data.

4 RESULTS AND DISCUSSION

This section reports the empirical results out-of-sample of the forecasting exercises conducted for the GDP of Rio Grande do Sul. Two exercises were performed, one using quarterly data and another using monthly data, each evaluated at short and longer horizons, 1 and 4 quarters, and 1 and 12 months ahead, respectively.

4.1 PREDICTIVE ACCURACY

Forecast accuracy was assessed through MAPE and rRMSE measures, while cumulative performance was analyzed using the CSFE (methodological details are in Section 2). To formally compare methods, the Diebold-Mariano test was applied, examining whether observed differences in predictive accuracy are statistically significant.

Tables 3 and 4 present MAPE and rRMSE for short- and long-term forecast horizons. In the monthly exercise MAPE does not exceed 1.5%, while in the quarterly exercise, MAPE remains below 5.5%. All alternative methods produce rRMSE values below 1 relative to the SARIMA benchmark, indicating more precision in terms of RMSE for the alternatives. Across horizons, LASSO obtained the lowest short-horizon errors one-month and one-quarter-ahead, while in the long-horizon, Random Forest was the method that achieved the lowest score, both in MAPE and rRMSE measurements. While these summary metrics provide static measures of accuracy, the CSFE plots that follow allow the view of forecast performance relative to SARIMA across time, in each horizon.

⁷An alternative approach considered for variables containing zeros or negative values was to shift the entire series upward by subtracting its minimum and adding a small constant, i.e., $(x - \min(x) + 2)$, before applying the logarithm. However, this transformation led to less precise forecasts overall and was therefore discarded.

⁸The grid for Elastic Net's λ consisted on a broad search of 200 values spaced on the log scale from 0.5 up to 25,000, while Random Forest's candidate *mtry* grid involved 1, $\sqrt{p}$, $p/3$, $p/2$, and p , with p being the number of predictors.

Table 3 – Mean absolute percentage error (MAPE, %) — Monthly and Quarterly settings

	Monthly analysis		Quarterly analysis	
	Horizon 1	Horizon 12	Horizon 1	Horizon 4
Benchmark	1.08%	1.48%	5.46%	5.09%
LASSO	0.59%	0.91%	3.19%	4.32%
Elastic Net	0.81%	1.08%	3.39%	4.34%
Random Forest	0.73%	0.85%	3.33%	4.10%

Source: Author's own elaboration.

Notes: Bold indicates the lowest error by column.

Table 4 – Relative RMSE (rRMSE) — Monthly and Quarterly settings

	Monthly analysis		Quarterly analysis	
	Horizon 1	Horizon 12	Horizon 1	Horizon 4
Benchmark	1.00	1.00	1.00	1.00
LASSO	0.53	0.60	0.53	0.69
Elastic Net	0.73	0.64	0.59	0.69
Random Forest	0.72	0.56	0.74	0.68

Source: Author's own elaboration.

Notes: Root mean squared error of methods relative to the SARIMA's in each forecast horizon. Bold indicates the lowest error by column.

Figures 2 and 3 display CSFE series, made of the difference between benchmark and method errors, for each forecasting horizon and method. An upward trend in a method's CSFE curve indicates accumulated superiority over the benchmark. In the quarterly exercise the machine learning methods generally accumulate lower squared errors than the SARIMA throughout the out-of-sample period, with a notable exception for Random Forest at Horizon, 4 between 3Q2018 and 2Q2020, where its CSFE temporarily declines. In the monthly exercise, ML methods typically show smaller accumulated squared errors than the benchmark. Random Forest underperforms briefly in Horizon 1 around the first months of 2020.

Figure 2 – CSFE – Forecasting horizons and methods – Monthly setting.

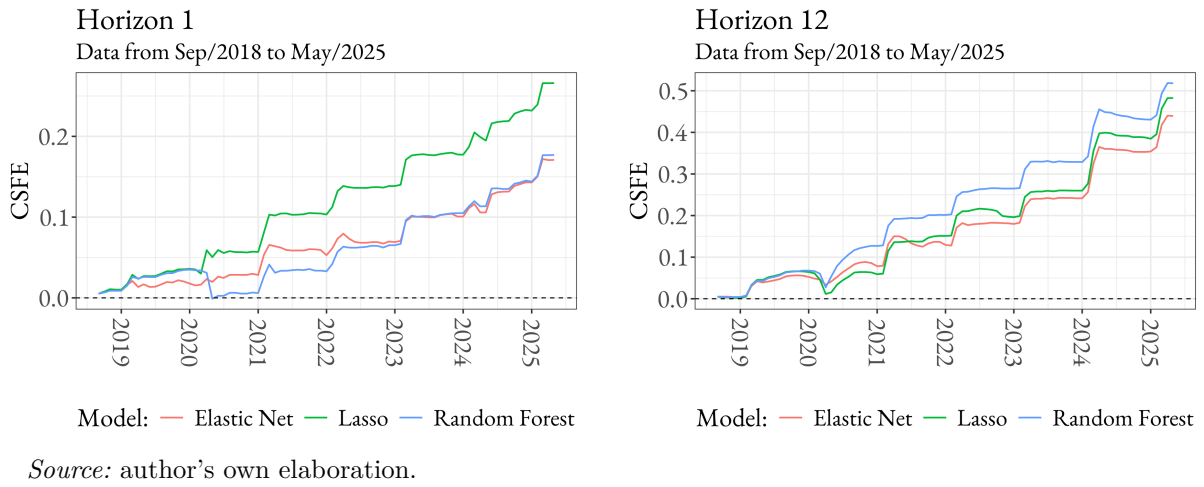
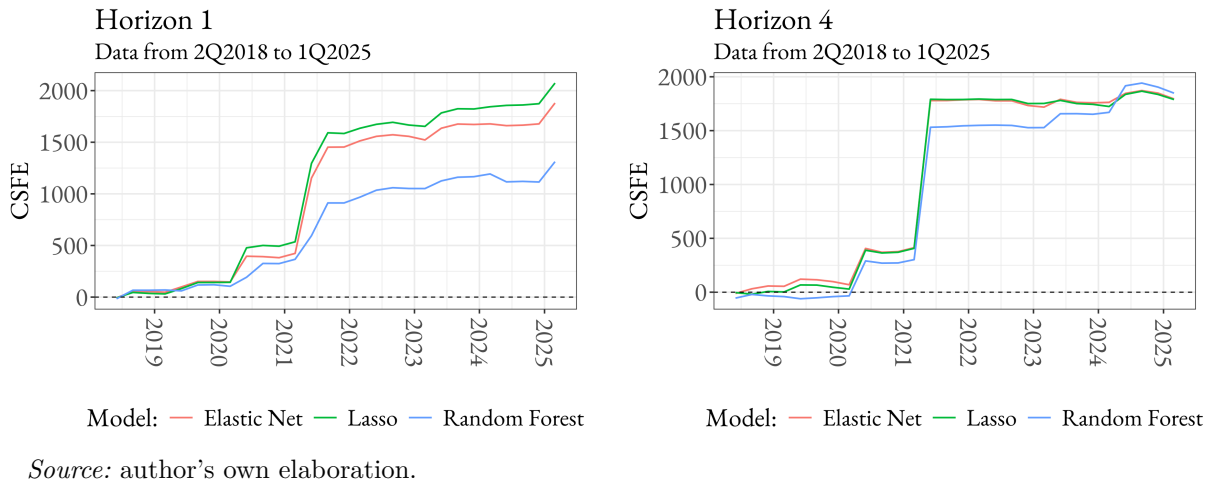


Figure 3 – CSFE - Forecasting horizons and methods - Quarterly setting.



To formally test whether these performance differences are statistically significant, the Diebold–Mariano (DM) test was applied. The test compares each method's predictive accuracy against the SARIMA benchmark, with the null hypothesis of equal predictive accuracy. Table 5 reports DM test p -values, and indicates that methods showed significant improvement altogether over the benchmark in all horizons, except for Horizon 4 in the quarterly analysis.

In the monthly analysis, all three machine learning methods rejected the null at the 1% level for Horizon 12. For Horizon 1, LASSO rejected it at the 1% level, while the other methods did so at the 5% level, indicating statistically significant improvements over SARIMA. In the quarterly analysis, the p -values are larger and, for Horizon 4, do not reach conventional significance levels. For Horizon 1, the Random Forest showed significance at the 5% level, and the other two methods, at the 10% level.

Table 5 – Diebold–Mariano Test p -values for comparison against Benchmark

	Monthly analysis		Quarterly analysis	
	Horizon 1	Horizon 12	Horizon 1	Horizon 4
LASSO	0.0003***	0.0007***	0.0218*	0.2291
Elastic Net	0.0031**	0.0001***	0.0281*	0.2186
Random Forest	0.0091**	0.0000***	0.0064**	0.1580

Source: author’s own elaboration.

Notes: Diebold-Mariano test p -values comparing each method against the benchmark.

Significance: 10% (*), 5% (**), and 1% (***).

4.2 DISCUSSION

The MAPE and rRMSE error metrics and the CSFE graphs are in concordance about the precision of the approaches. On average, the machine learning alternatives improve forecast accuracy relative to the SARIMA benchmark across most horizons and frequencies: in the monthly setting LASSO produces the lowest MAPE and rRMSE at Horizon 1, and Random Forest gives the lowest MAPE and rRMSE at Horizon 12. This pattern of method performance is also noticeable in the quarterly short- and long-term horizons. These aggregate measures confirm that the alternative methods reduce both absolute and squared forecast errors relative to the benchmark.

However, the CSFE plots reveal that these improvements are not uniform over time. A CSFE graph shows the accumulated difference between the squared errors of the benchmark and the method, therefore, the trend of the lines in the graph are directly correlated to their superiority in accuracy against the benchmark. The strong upward jumps seen in both short-term and long-term horizon graphs in Figure 3 indicate that the machine learning methods substantially outperformed SARIMA around the first quarter of 2021. This suggests that the structural changes in the volatile period immediately after the Covid-19 pandemic were challenging for the benchmark to adapt to.

This effect is not as evident in the monthly frequency exercise, but instead, in Figure 2 the recurring “staircase” shape increasing at the start of each year further implies that the SARIMA underestimates the magnitude of the seasonal spikes in GDP, making the forecasts of the alternative methods significantly more precise in comparison.

The Diebold–Mariano test in Table 5 provides statistical validation for these findings. In the monthly setting the p -values are strongly significant across horizons, with all methods presenting significance at 5% or less for Horizon 1 and at 1% for Horizon 12, results that indicate the performance improvements observed in MAPE and rRMSE are not due to random variation.

In contrast, the quarterly results are mixed: methods achieve statistically significant improvements at the short horizon, with Random Forest presenting significance at 5%, LASSO and Elastic Net at 10%, but fail to show significance at the longer quarterly horizon, where all p -values are bigger than 0.15. Therefore, while the monthly improvements are robust both economically and statistically, the quarterly gains are weaker and less stable at longer horizons. This frequency dependence suggests that higher temporal resolution helps the ML methods capture patterns and shocks, that are smoothed over in quarterly aggregation.

An intriguing detail emerges in the quarterly Horizon 1 results. Despite LASSO achieving the smallest MAPE and rRMSE errors, its DM test significance is weaker (10%) than that of the Random Forest (5%). This apparent inconsistency reflects the nature of the Diebold–Mariano test, which evaluates not only the average magnitude of error improvements but also their temporal consistency. LASSO’s gains over the benchmark may be concentrated in specific episodes, while Random Forest delivers more stable, though slightly smaller, improvements across time. As a result, the Random Forest demonstrates stronger statistical evidence of superiority despite a marginally higher average error.

The findings of this study agree with previous literature. For example, the work conducted by Medeiros and Vasconcelos (2016) and Cepni, Güney, and Swanson (2019), when revealing that sparsity methods such as the LASSO yield very accurate forecasts in short-term horizons against benchmark autoregressive models when estimating Brazilian GDP. Also, go in accordance to Jokubaitis, Celov, and Leipus (2021), when in a quarterly setting pointed LASSO outperformed the benchmark ARMA for U.S. and EU GDP forecasts, particularly during crisis periods.

Moreover, Yoon (2021) found Random Forest, together with Gradient Tree Boosting, to be the best performing methods in the prediction of Japan’s GDP, especially at longer horizons. Similarly, for U.S. inflation, Boesch and Ziegelmann (2025) report results in line with this work’s contrast between short- and long-term accuracy: shrinkage methods achieve their strongest gains in short-horizon settings, whereas RF and other tree-based methods deliver more stable accuracy across horizons, performing especially well during volatile periods, such as the pandemic. Araujo and Gaglianone (2023) also concluded Random Forest to be among the two best methods for longer horizons in forecasting Brazil’s monthly inflation, according to the DM test performed in their work. In contrast, they reported weak results for LASSO and Elastic Net when compared to the other models tested.

Overall, evidence points out that machine learning methods show better predictive accuracy than the benchmark, although not consistently throughout time, forecast horizons and/or frequencies. This suggests the performance of the algorithms is strongly dependent on data availability and the underlying relationships between variables, implying that the use of machine learning together with traditional time series models might be the most propitious scenario for forecasts across different settings.

5 CONCLUSION

The purpose of this study is to evaluate the precision of forecasting regional activity of three machine learning methods, LASSO, Elastic Net and Random Forest, and compare their performance with that of a traditional econometric model, SARIMA. The target variable is the GDP of Rio Grande do Sul (RS), Brazil’s southernmost federative state. Out-of-sample predictions between 2018 and 2025 are produced using a direct forecasting approach, with separate models for each horizon, and the exercise is conducted at two frequencies, monthly (1- and 12-steps-ahead) and quarterly (1- and 4-steps-ahead).

Results show that all machine learning methods achieved statistically significant improvements over the SARIMA benchmark across horizons, according to the Diebold-Mariano test, with the exception of Horizon 4 in the quarterly setting. The LASSO is the most accurate method for short-term horizon forecasts, while for the long term, the Random Forest showed the greatest precision, in both monthly (where the two obtained significance at the 1% level for their respective horizons) and quarterly frequencies.

The empirical contribution of this study comes in two ways. First, it develops predictive models for GDP forecasting that incorporate macroeconomic and weather data. Second, it fills an important research gap by generating localized insights into the economy of RS, which is significant, but volatile due to climatic and cyclical factors. By doing so at two different frequencies, it exploits complementary advantages of each. Monthly data contain higher-frequency indicators that provide more information to the models, while quarterly data align with official GDP releases; at the same time, the two frequencies allow testing the ML methods across relatively data-rich and data-scarce environments.

It is necessary to investigate whether machine learning methods can improve the accuracy of regional GDP forecasts, especially in subnational economies that receive limited research attention. Enhanced accuracy could enable state-level policymakers to allocate resources efficiently, anticipate economic cycles, and intervene more effectively in case of floods or commodity shocks. It would also help businesses and investors make more informed decisions about production, hiring, and investment, especially given the increased volatility of emerging economies.

Despite these positive results, the study contains noteworthy limitations. Even though the chosen machine learning methods account for high-dimensional data, they can still exhibit instability in feature importance and risk overfitting. Complementing the ML methods with variable selection algorithms, similar to the approach of Barbosa, Ferreira, and Silva (2020), could help mitigate these risks. Furthermore, aggregations in the quarterly exercise may hinder the identification of important relationships between variables. Therefore, incorporating mixed-data sampling (MIDAS) regressions, as in Chu and Qureshi (2023), is a possibility worth exploring.

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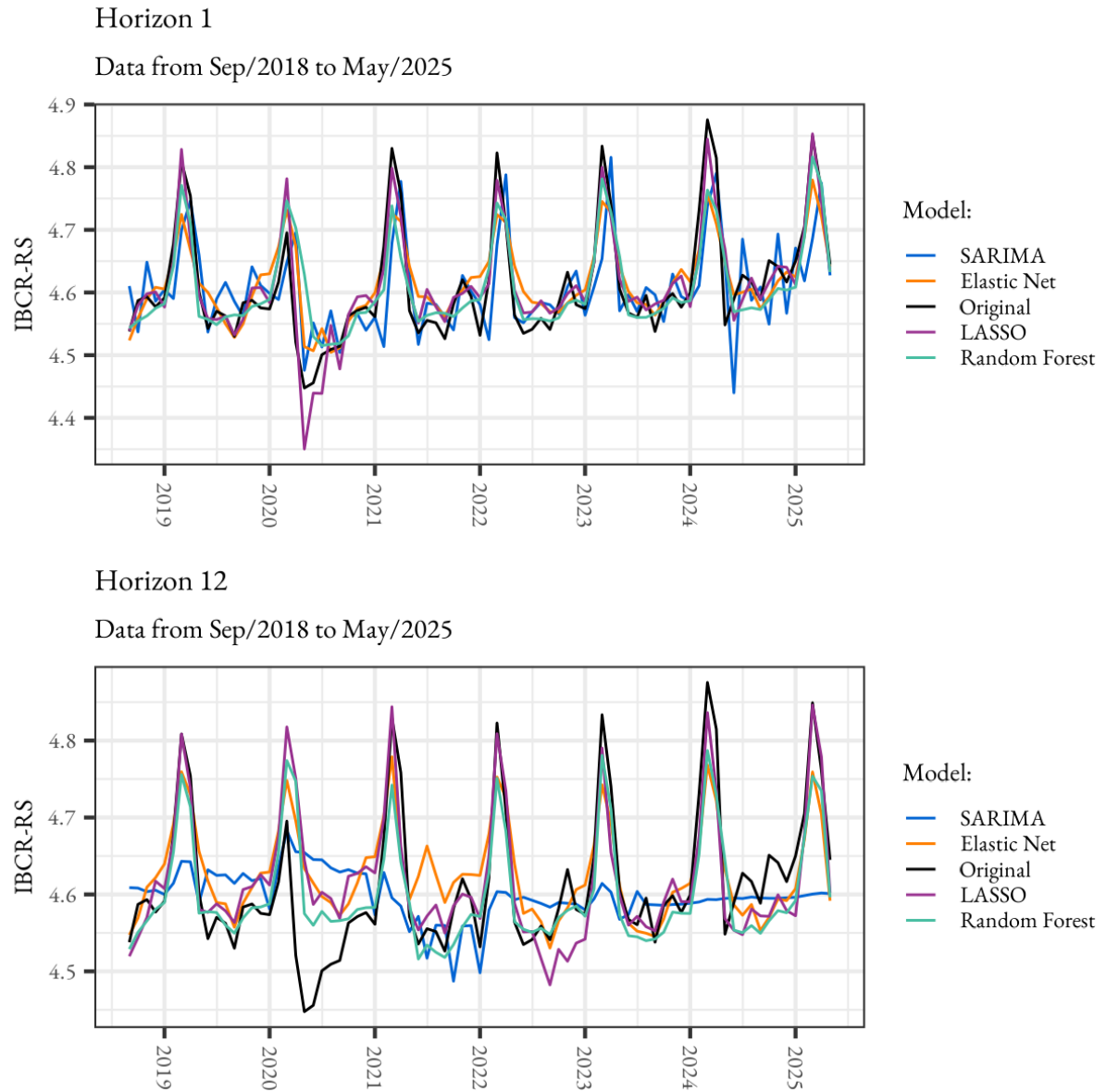
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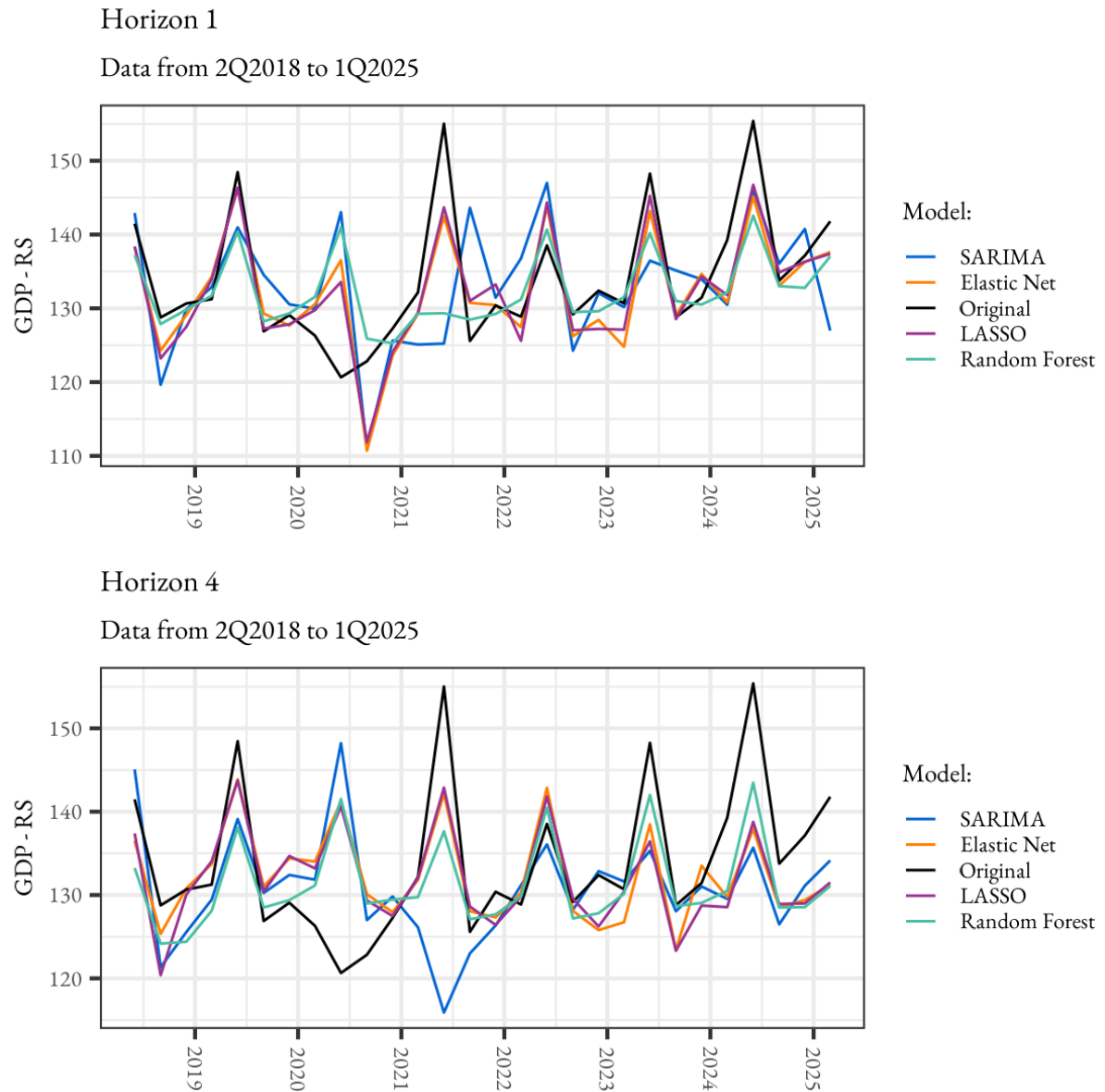
APPENDIX A - FORECAST GRAPHS AND RMSE VALUE

Figure 4 – Forecast results – Monthly setting.



Source: author's own elaboration.

Figure 5 – Forecast results – Quarterly setting.



Source: author's own elaboration.

Table 6 – Root mean squared error (RMSE) — Monthly and Quarterly settings

	Monthly analysis		Quarterly analysis	
	Horizon 1	Horizon 12	Horizon 1	Horizon 4
Benchmark	0.067	0.096	10.14	11.06
LASSO	0.035	0.058	5.36	7.65
Elastic Net	0.049	0.062	5.97	7.63
Random Forest	0.049	0.054	7.48	7.51

Source: Author's own elaboration.

Notes: Root mean squared error of models in each forecast horizon. Bold indicates the lowest error by column.

APPENDIX B - APPROVAL SHEET

EDUARDA TEIXEIRA SAYAGO

STATE GDP FORECASTING WITH MACHINE LEARNING: A CASE STUDY OF RIO GRANDE DO SUL

Trabalho de conclusão submetido ao
Curso de Graduação em Ciências Econômicas
da Faculdade de Ciências Econômicas da
UFRGS, como requisito parcial para obtenção
do título Bacharel em Economia.

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BANCA EXAMINADORA:

Prof. Dr. Marcelo Savino Portugal

Orientador - UFRGS

Prof. Dr. Flávio Augusto Ziegelmann

UFRGS

Prof. Dr. Sabino da Silva Porto Júnior

UFRGS